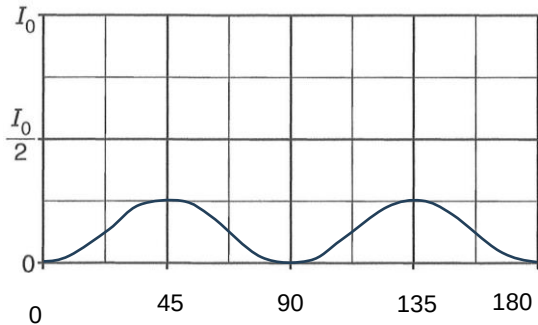


Qns	Answer	Marks
1a	 Correct positions of max and min intensities, and gradient at 0° and 180° should be approx zero [B1] Max intensity of $I_0/4$ [B1] Working: $I_\theta = I_0 \cos^2\theta \cos^2(90-\theta)$ $= I_0 \cos^2\theta \sin^2\theta$ $= I_0/4 \sin^2 2\theta$ [note: $\sin 2\theta = 2\sin\theta \cos\theta$]	
1bi	$d = 10 \times 10^{-6} \div 7$ $= 1.43 \times 10^{-6} \text{ m}$	M1 A1
1bii	For number of images on each side, $\sin\theta = 45^\circ$ $n_{\max} = d \sin 45^\circ / \lambda$ $= 1.43 \times 10^{-6} \times \sin 45^\circ / 600 \times 10^{-9}$ $= 1.68$ ≈ 1 Total number of images = $1 + 1 + 1 = 3$	C1 A1
1biii	For the first order maxima, $d \sin \theta = \lambda$. <u>Most of the line spacing in the grating is uniform</u> such that the rays converge at α to give a bright image. <u>Some of the line spacing is smaller than the standard spacing</u>, giving rise to some part of the image being formed at a larger angle between α and β.	B1 B1

1ci	The stationary wave is formed when the incident wave is <u>reflected at the sea wall</u> such that the incident and reflected <u>waves are of the same type, amplitude, frequency, wavelength and speed</u> the incident and reflected wave travelling in <u>opposite directions then overlap and superpose.</u>	B1 B1
1cii	For the increase in amplitude at wall, the incident and reflected waves must have <u>amplitudes of the same sign / crest meets with crest.</u> OR Constructive interference The reflected wave must have the <u>same phase</u> as the incident wave, hence <u>no phase change</u> occurred during reflection.	M1 A1
1ciii	$1.75\lambda = 0.78$ $\lambda = 0.78/1.75$ $v = f\lambda$ $= 720 \times 0.78/1.75$ $= 321$	C1 A1

Qns	Answer	Marks
2(cii)	The increase in internal energy of a system is the	D1